B.Tech (ME)– IV-I Sem

23A03702e	POWER PLANT ENGINEERING (Professional Elective-IV).	L	T	P	C
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Course objectives: The objectives of the course are to	
1	Familiarize the sources of energy, power plant economics and environmental aspects.
2	Outline the working components of different power plant.
3	Understand the working mechanism of diesel and gas turbine power plants.
4	Impart types of nuclear power plants, and outline working principle and advantages and hazards.
5	Explain renewable energy sources; characteristics, working principle, classify types, layouts, and plant operations.

Course Outcomes: On successful completion of the course, the student will be able to,		
1	Outline sources of energy, power plant economics, and environmental aspects.	L2
2	Describe working components of a steam power plant.	L2
3	Illustrate the working mechanism of diesel and gas turbine power plants.	L2,
4	Summarize types of renewable energy sources and their working principle.	L2
5	Demonstrate the working principle of nuclear power plants.	L4

UNIT I

Introduction to the Sources Of Energy - Resources and Development of Power in India. Convectional and non- conventional energy sources, Power Plant Economics and Environmental Considerations: Capital Cost, Investment of Fixed Charges, Operating Costs, General Arrangement of Power Distribution, Load Curves, Load Duration Curve. Definitions of Connected Load, Maximum Demand, Demand Factor, Average Load, Load Factor, Diversity Factor - Tariff - Related Exercises. Effluents from Power Plants and Impact on Environment - Pollutants and Pollution Standards - Methods of Pollution Control. Inspection And Safety Regulations.

UNIT II

Steam Power Plant : Introduction to Boilers- Modern High Pressure and Supercritical Boilers - Analysis of Power Plant Cycles - Modern Trends in Cycle Improvement - Waste Heat Recovery, Fluidized Bed Boilers., Fuel and Handling Equipments, Types of Coals, Coal Handling, Choice of Handling Equipment, Coal Storage, Ash Handling Systems.

Steam Power Plant : Combustion Process : Properties of Coal - Overfeed and Under Feed Fuel Beds, Travelling Grate Stokers, Spreader Stokers, Retort Stokers, Pulverized Fuel Burning System And Its Components, Combustion Needs and Draught System, Cyclone Furnace, Design and Construction, Dust Collectors, Cooling Towers And Heat Rejection. Analysis of Pollution from Thermal Power Plants - Pollution Controls.CO2 Recorders

UNIT III

Diesel Power Plant: Diesel Power Plant, Construction, Plant lay out with auxiliaries, fuel storage.
GAS TURBINE PLANT: Introduction - Classification - Construction - Layout with Auxiliaries - Principles of Working Closed and Open Cycle Gas Turbines. Advantages And Disadvantages Combined Cycle Power Plants.

UNIT IV

Hydro Electric Power Plant: Water Power - Hydrological Cycle / Flow Measurement - Drainage Area Characteristics - Hydrographs - Storage and Pondage - Classification of Dams and Spill Ways. Hydro Projects Plant: Types - Typical Layouts - Plant Auxiliaries - Plant Operation Pumped Storage Plants.

UNIT V

Power from Non-Conventional Sources: Utilization of Solar Collectors- Working Principle, Wind Energy - Types of Turbines - HAWT & VAWT-Tidal Energy. MHD power Generation.
Nuclear Power Station: Nuclear Fuel - Nuclear Fission, Chain Reaction, Breeding and Fertile Materials - Nuclear Reactor -Reactor Operation.

Types of Reactors: Pressurized Water Reactor, Boiling Water Reactor, Sodium-Graphite Reactor, Fast breeder Reactor, Homogeneous Reactor, Gas Cooled Reactor, Radiation Hazards and Shielding - Radioactive Waste Disposal.

TEXT BOOKS:

1. P.K. Nag, “Power Plant Engineering”, 3rd edition, TMH, 2013.
2. Wakil, “Power plant technology”, M.M.EI TMH Publications.

REFERENCE BOOKS:

1. Rajput, “A Text Book of Power Plant Engineering:, 4th edition, Laxmi Publications, 2012.
2. Ramalingam, “Power plant Engineering”, Scietech Publishers, 2013
3. P.C. Sharma, “Power Plant Engineering”, S.K. Kataria Publications, 2012.
4. Arora and S.Domakundwar, “A course in Power Plant Engineering”, Dhanpat Rai & Co (p) Ltd, 2014.

Online Learning Resources:

https://nptel.ac.in/courses/112107291

https://youtube.com/playlist?list=PLLy_2iUCG87BT8H9uMufjrcPF5e6Qd2bz&si=RQhZwEibgqXK2dRL

https://youtube.com/playlist?list=PLLy_2iUCG87BT8H9uMufjrcPF5e6Qd2bz&si=LjgzdabT7tIsCwJC

https://youtu.be/IdPTuwKEfmA?si=PyF04z9beiVGkXAS